

IBM Data Analyst Capstone Project

Richard Edwards
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OUTLINE



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EXECUTIVE SUMMARY



Overall the Stack Overflow Survey shows a few key takeaways and trends:

- Web and Enterprise technologies are the highest in demand.
- That there are many correlations between respondent demographics
 - More experienced developers generally have higher job satisfaction.
 - Years of experience also has a positive relationship to earning potential.
- For databases many respondents seem to prefer relational databases like MySQL and PostgreSQL



INTRODUCTION



The purpose of this report is to breakdown and analyze the Stack Overflow User Survey

- The intended audience is developers and employers in the IT and data fields.
- The value of analyzing this data is:
 - To understand current trends of the platforms and tech stacks that are popular for use.
 - To predict what future trends may emerge based on the tech that developers would prefer to be using.

METHODOLOGY

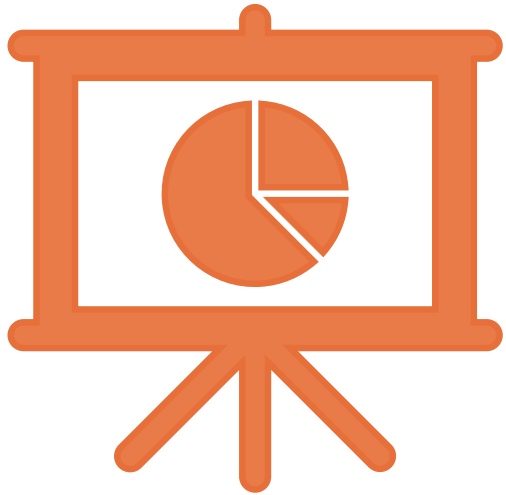


Data sources, Data collection & Data wrangling steps

- The data source for this project was the Stack Overflow Developer Survey.
- We also scraped data from the web to corroborate on some of the trends shown for popular coding languages.
- The data was wrangled using several steps including:
 - Removing missing values by imputing the mean, median, or mode, or dropping the values when they were insignificant.
 - Normalizing categorical data to eliminate skew and make the data more precise.



Results



The Results can be seen in the following slides:

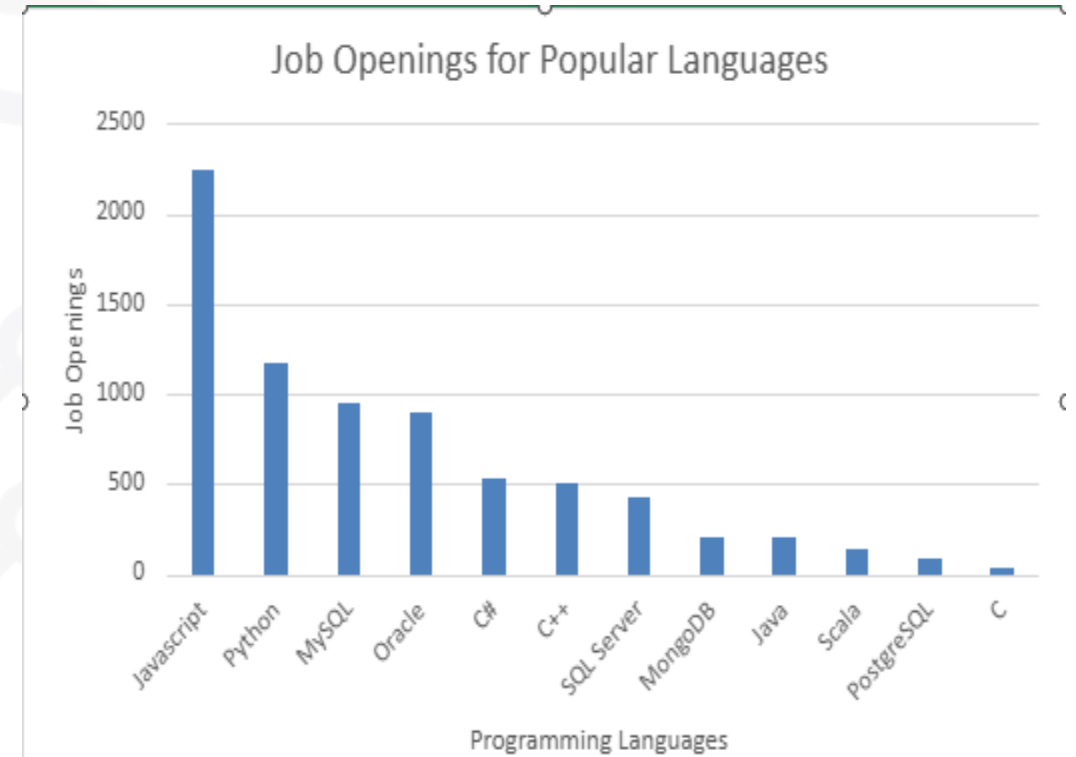


JOB POSTINGS

Insights:

-Javascript, Python, and MySQL are definitely the most in demand skills based on job listings while older and more niche languages are not as in demand.

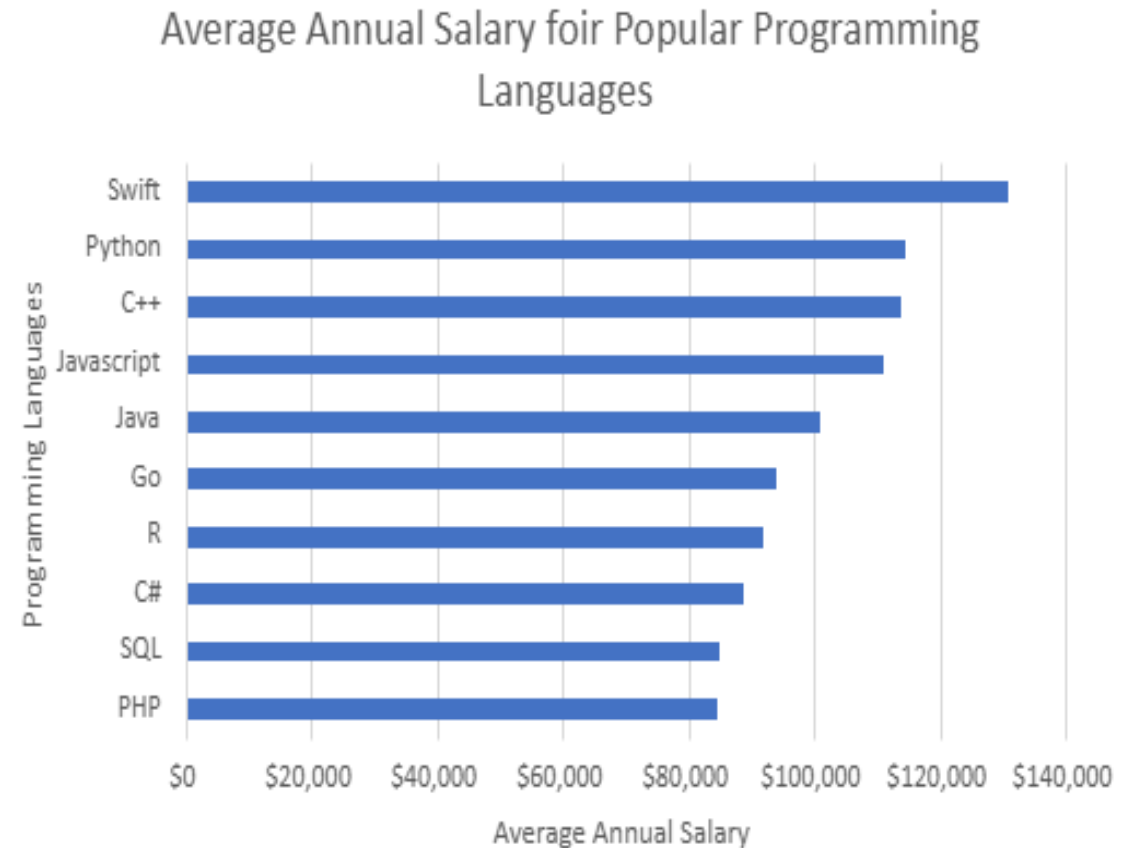
-Having a good mix of database and web skills should help with finding a career in technology.



POPULAR LANGUAGES

Insights:

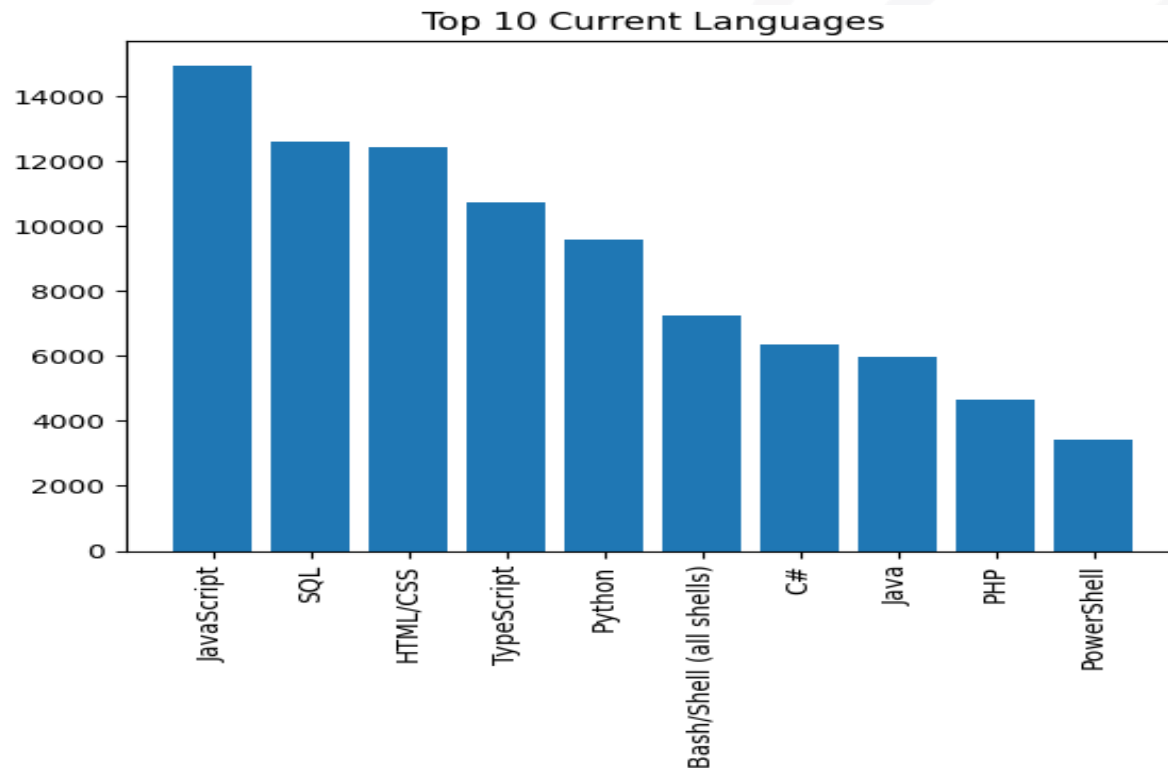
- While all languages typically provide higher salaries when compared to the national average some typically demand high than others.
- The Languages that pay higher are typically either web, enterprise, or platform specific languages.



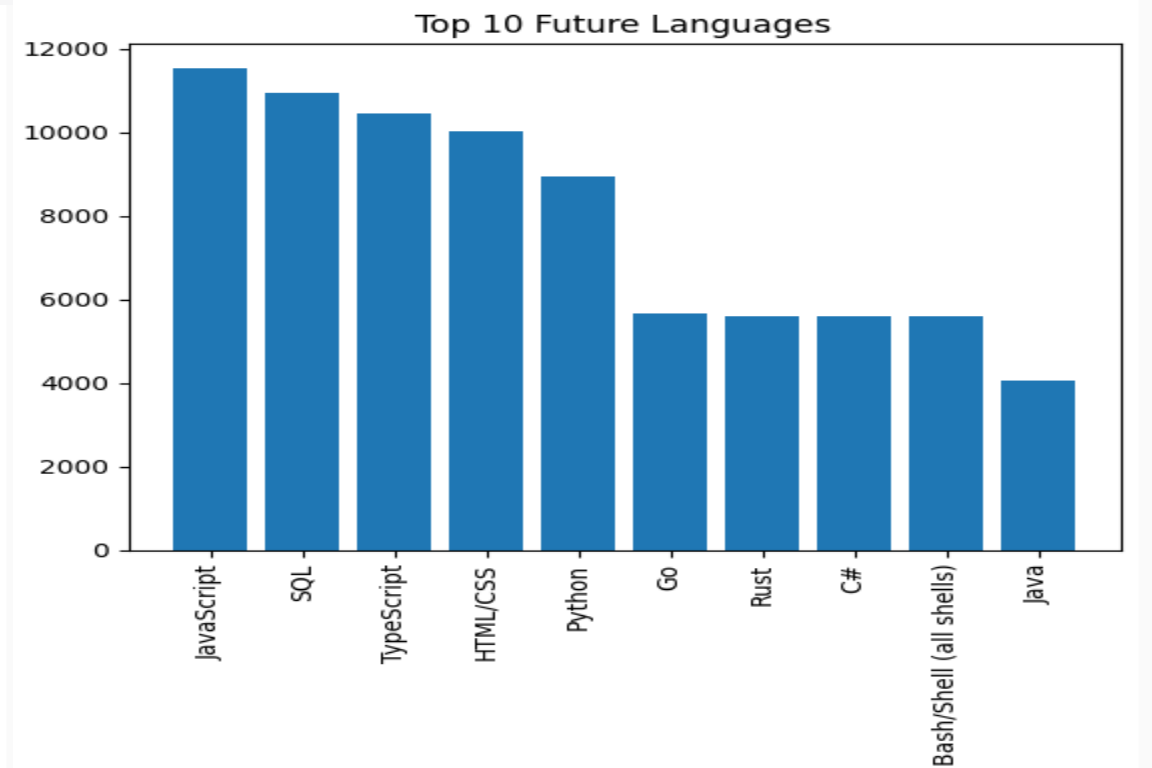
PROGRAMMING LANGUAGE TRENDS

Based on these charts Web-languages and Python are currently the most common and the most desired.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Python, SQL, JavaScript/Typescript, and HTML/CSS continue to be the most popular and in demand languages.
- C-Based and platform-specific languages can demand higher salaries on average.

Implications

- Web and Enterprise have strong futures and job growth and are a safe bet to learn and grow with.
- Understanding a niche or domain specific programming language could help secure higher levels of employment.

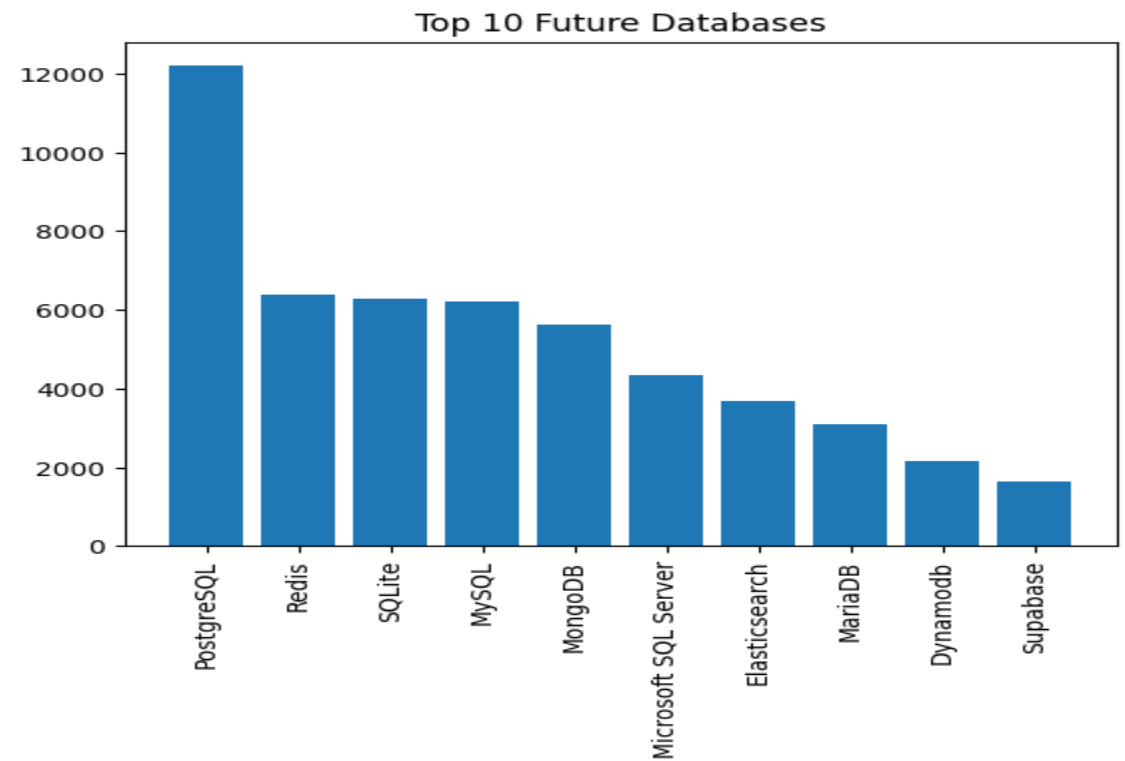
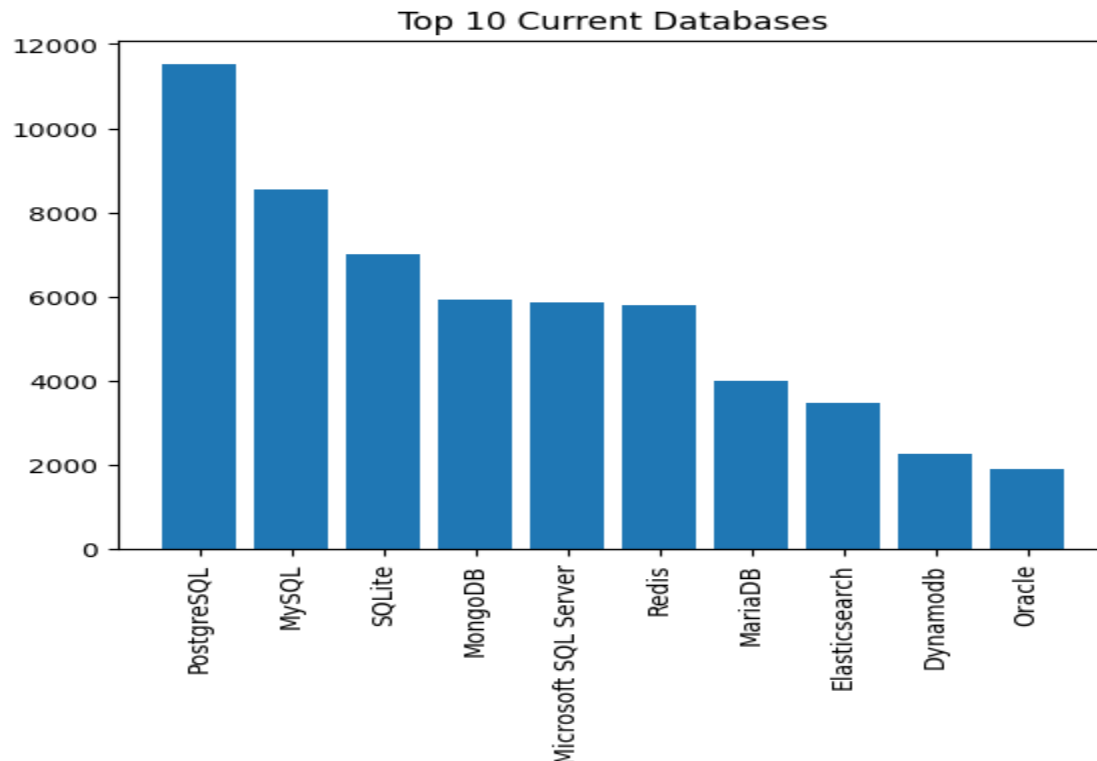


DATABASE TRENDS

Overwhelmingly PostgreSQL is the most popular and looks to continue that trend into next year, other relational databases are also quite popular.

Current Year

Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most popular current and future database to use.
- Relational databases in general seem to be preferred over other types.
- MongoDB and Redis are both relatively popular as non-relational options.

Implications

- Understanding how to use PostgreSQL seems the most important for finding employment as a data analyst.
- Other relational databases are still worth knowing and understanding.
- The upward trend of Redis in popularity could indicate that non-relational databases are becoming more common.



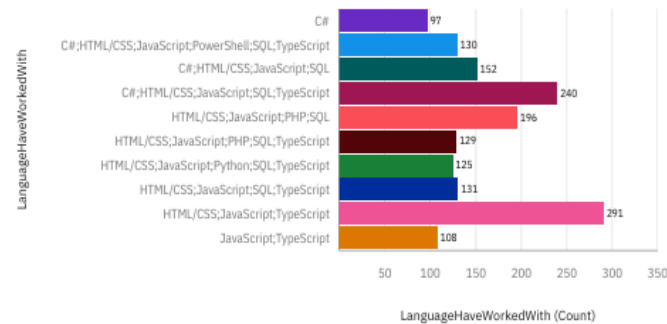
DASHBOARD



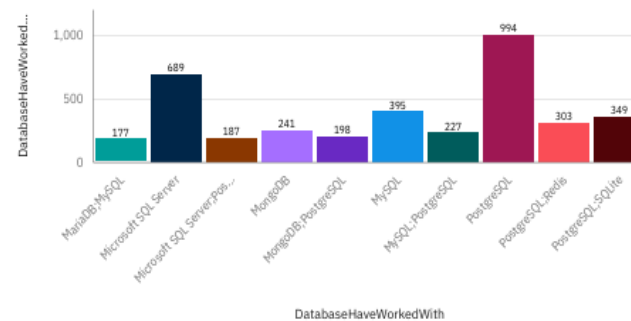
DASHBOARD TAB 1: Current Technology Usage

Current Technology Usage

Top 10 Languages Used



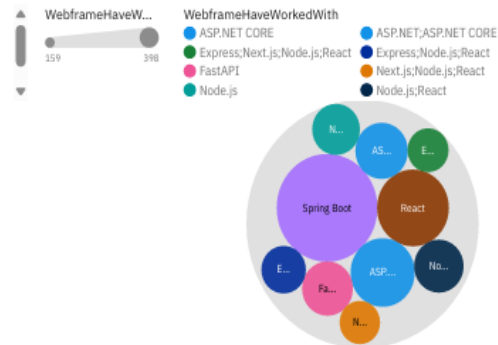
Top 10 Databases Used



Top 10 Platforms Used



Top 10 Web Frameworks Used



This dashboard highlights the current popular technologies based on the survey data.

- The key insights are that developers typically know multiple languages, and web and enterprise languages are dominant.
- Additionally, AWS and Azure dominate for platform usage, and Spring Boot and React are the most popular web frameworks.

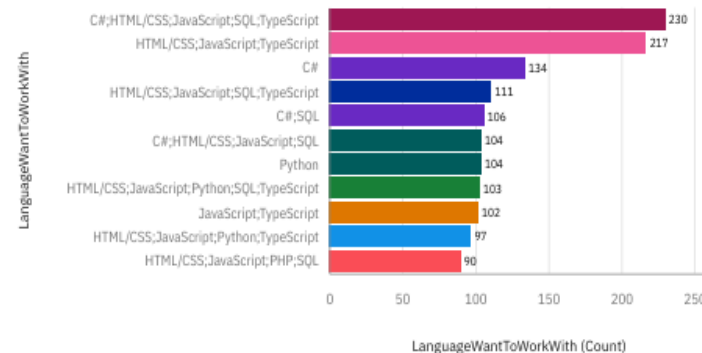
DASHBOARD TAB 2: Future Technology Trends

This dashboard highlights future trends for different tech stacks.

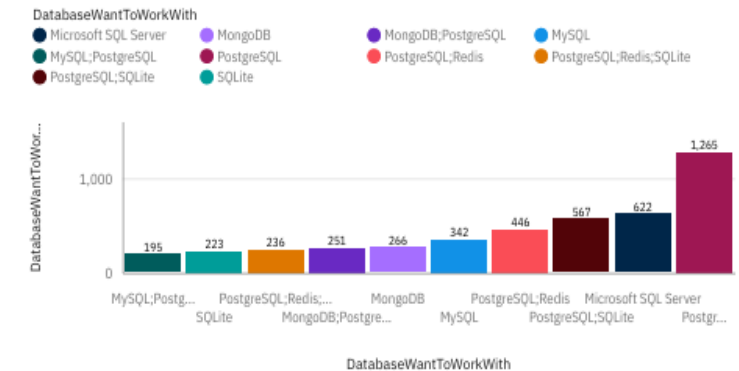
- Based on this web and enterprise languages such as C# and JavaScript seem to be what people want to work with more, as well as PostgreSQL is the preferred database.
- AWS and Azure are overwhelmingly the platforms that developers want to use, while spring boot and React remain the web frameworks of choice, with ASP.NET Core rounding out as the third choice.

Future Technology Trend

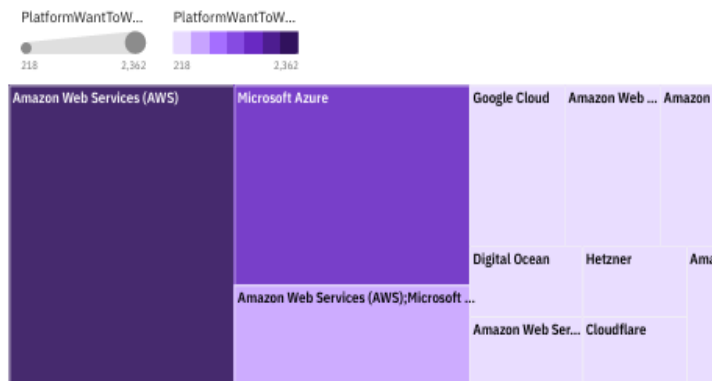
Top 10 Languages People Want to Work With



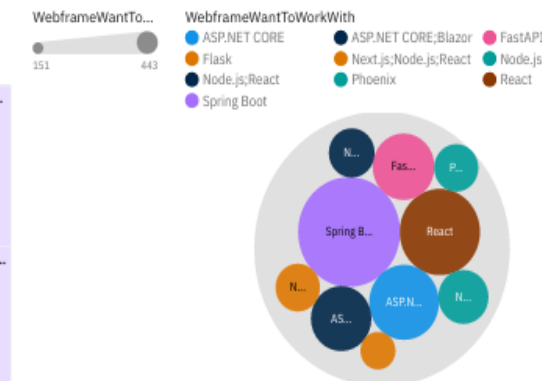
Top 10 Databases Developers Would Prefer to Work With



Top 10 Platforms Developers want to Work With



Top 10 Web Frameworks that People Want to Use



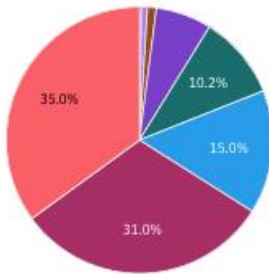
DASHBOARD TAB 3: Demographics

Demographics

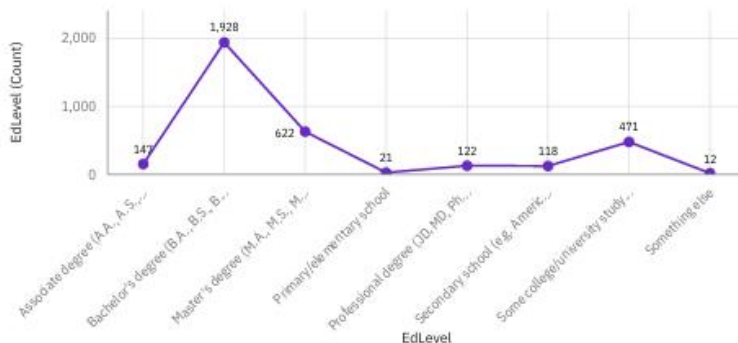
Respondents by Age

Age

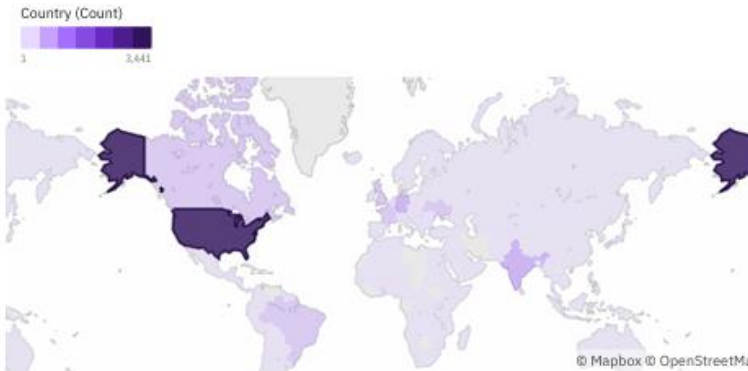
- Prefer not to say
- Under 18 years old
- 65 years or older
- 55-64 years old
- 18-24 years old
- 45-54 years old
- 35-44 years old
- 25-34 years old



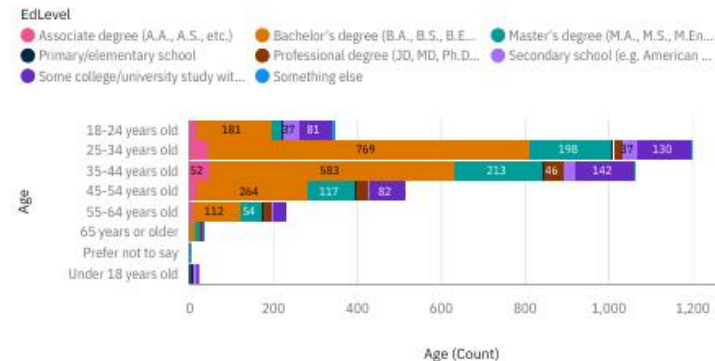
Respondent Count by Education Level



Respondent Count by Country



Respondent Count by Age Classified by Education Level



This highlights some of the demographics of survey respondents

- Most respondents are between the ages of 25 to 45 and hold a Bachelor's degree.
- Looking by Country the US has the most respondents followed by India, Brazil, Canada, and several EU nations
- Beside bachelor degrees, many of the respondents either have master's degrees or partial college study.

DISCUSSION



- Overall, the trends in tech point towards a continued dominance of web and enterprise technologies.
- Additionally, most developers seem to be between the ages of 25-45 and have at least a bachelor's degree.
- Amazon and Microsoft overwhelmingly dominate that platform space with AWS and Azure respectfully.



OVERALL FINDINGS & IMPLICATIONS

Findings

- Python and JavaScript are overwhelmingly the most popular programming languages
- PostgreSQL dominates for database usage.
- Most developers are in their 20s and 30s.

Implications

- Learning these technologies is very important for finding employment.
- Relational databases continue to be the preferred option for data storage and manipulation.
- Many developers still have lots of space to grow within their careers.



CONCLUSION



In conclusion, there are many great and growing programming languages that are valuable to learn.

And while there are new languages and platforms arising in the tech space, web and enterprise tech stacks are still the most prevalent.



APPENDIX



- Include any relevant additional charts, or tables that you may have created during the analysis phase.

